

Methods — bullet outline for the manuscript

Untracked working document. Every entry is a fact pulled from the code, the README or the run records, with the source named where it is not obvious. Nothing here is drafted prose.

Numbers in `code` font are defaults as coded; where a run used something other than the default it is said so. Items marked **[decide]** are choices only you can make; **[gap]** marks something the record does not currently contain.

1. Ligand and system definition

- Ligand: built-in `octinoxate` entry, **C17H24O3** — one CH₂ short of real octinoxate (2-ethylhexyl 4-methoxycinnamate, C₁₈H₂₆O₃). SMILES perceived from the stored geometry: `CCCC[C@H](CC)OC(=O)/C=C/c1ccc(OC)cc1`; the ester oxygen sits on the chain CH. (`NEXT_STEPS.md`, written 2026-09-24.)
- [decide]** whether the paper calls this an octinoxate analogue throughout, or re-runs on the true C18 molecule. Every number in the project is for the C17 analogue.
- Ligand net charge 0, singlet; 20 heavy atoms (the denominator of every "engaged x/20" count).
- Binding-site box: ligand bounding box padded by `--box-pad` (default 0.0 Å); poses are sampled uniformly inside it.

2. Fragment library

- Ten side-chain **analogues**, each capped where the backbone would attach — not residues. Isoleucine entry is n-butane, leucine entry is isobutane; neither carries its namesake branch. Tyrosine is p-cresol, glutamic is propionate; all ten are chemically distinct.
- One line needed for reproducibility: two entries differ from the upstream library they came from (tyrosine and glutamic were wrong there), so results are not comparable with runs made against it. `attach_geom.py` checks electron parity, so a radical declared closed-shell cannot pass.
- `attach_geom.pose_attachment` locates each fragment's capping carbon (the CB marker) and the positions a CA can be grafted onto it.

3. Pose placement and scoring

- Placement: random position inside the padded box, random rotation about x, y, z (`get_frag_coordinates`), `--tries 100` attempts per fragment.
- Acceptance: unrelaxed interaction energy below `--energy-cutoff 500` kcal/mol, then BFGS relaxation of the fragment with all ligand atoms fixed (`FixAtoms`), `fmax 0.05` eV/Å, `steps 200`.
- Interaction energy $IE = E(\text{complex}) - E(\text{ligand}) - E(\text{fragment})$, all at the relaxed pose geometry, converted with 23.06035 kcal/mol per eV.
- Potential: **FAIRChem UMA**, checkpoint `uma-s-1p2p1`, `FAIRChemCalculator(task_name="omol")` via ASE (`peptide_builder.py:1079`). CPU only on Apple Silicon (FAIRChem accepts `cpu/cuda`, no MPS). Charge and spin are passed per subsystem in `atoms.info`; totals combine as `charge_A + charge_B` and `spin_A + spin_B - 1`.
- Yield for octinoxate: **165 accepted poses across ten side-chain types**.
- Pose-class energetics, needed later for the charge-bias argument: charged analogues average **-10.79**, aromatic **-4.71**, neutral aliphatic **-2.94** kcal/mol.

4. Shell selection

- Greedy and order-dependent: walk the fragments in a given order, each taking its lowest-IE pose that does not come within `--clash 1.4` Å of any pose already selected; repeated `--copies` times so a fragment may be placed more than once.
- Rotating which fragment picks first (`--all-starts`) × {1, 2} copies → **20 shells**, 8–13 fragments, drawing on 48 distinct poses.

- Independence caveat that belongs in Methods, not just Limitations: only **18 of the 20 are distinct**, and **9 of those are strict subsets** of another (each single-copy shell is the first pass of its two-copy counterpart; the lysine and aspartic walks converge). → nine independent arrangements plus nested subsets.
- Three shells swept and folded in full:

	shell 1	shell 2	shell 3
seed / selection	richest	second richest	tryptophan pose 2, chosen deliberately weak
total fragment IE (kcal/mol)	-113.12	-109.15	-60.40
charged poses	7 (58%)	5 (42%)	4 (33%)
aromatic poses	1 (8%)	2 (17%)	4 (33%)
shared poses with shell 1	—	5 / 12	fraction

- [decide]** whether shell 3's selection (an a-priori test of the selection criterion) is presented in Methods as a designed experiment or in Results as a follow-up.

5. Reachability measurement (the step that distinguishes the method)

- What is bridged is **attachment point to attachment point**, not centroid to centroid — the two are several Å apart for a large residue like tryptophan, and only the attachment points carry a direction.
- Construction: hold two poses fixed, graft a CA onto each capping carbon, grow an **ideal trans backbone** from one CA to the other through k glycine spacers by NeRF.
- Ideal geometry held fixed (Å, degrees): N-CA 1.458, CA-C 1.525, C-N 1.329, C=O 1.231, CA-CB 1.530, N-H 1.010, CA-HA 1.090; N-CA-C 111.2, CA-C-N 116.2, C-N-CA 121.7, CA-C-O 120.8, CB-CA-C 110.5, CB-CA-N 110.5, C-N-H 119.0; **omega fixed at 180°**.
- Free parameters: τ_1 (rotation of residue A's tripod about CB-CA), τ_2 (psi of A, shifted by the CB reference), and ϕ/ψ per spacer → **2 + 2k**.
- Closure condition: residue B's amide N must land on the circle compatible with B's fixed CA and CB — a 111° cone of radius 1.458 Å — i.e. two conditions; residual reported in Å.
- Optimisation: `scipy.optimize.minimize`, L-BFGS-B, `--restarts 24` random starts, `--seed 0`.
- Optional steric term (`--steric`) penalising backbone–ligand and backbone–fragment overlap.
- Sweep: every **ordered** pair at $k = 0 \dots 4$ (`--kmin 0 --kmax 4`); 12 poses → 132 ordered pairs.
- Tolerance used downstream: closure < `--tol 0.2` Å counts as connectable (README's rationale: absorbable by real backbone flexibility).
- Measured results to report as method validation:
 - k = 0 closed in 0 of 810 attempts**; 2% even at a deliberately generous 1.0 Å tolerance.
 - At $k = 2$, a partner within 30° of the residue's CA→CB direction reaches **7.5 Å**; a partner approaching from the side reaches **11.1 Å**.

6. The spacer budget

- Counting argument: for a path through m steps the free dihedrals number $1 + m + 2 \cdot \sum k$ against $4m$ closure conditions → generically closable when $\sum k \geq (3m - 1)/2$.
- Stated as a **global** budget, not a per-step test: surplus in one segment pays a shortfall in the next.
- Criterion is **necessary, not sufficient** — degrees of freedom plus a steric term; one three-fragment path that met its budget closed while another did not.

7. Path assignment

- Nodes = placed poses (worth their IE); edge $a \rightarrow b$ exists only if the sweep found a connection under tolerance; edge cost = k spacers + residue b .
- Exact** search: Held–Karp dynamic program over (visited set, current node) — $2^n n$, no greedy walk. `--lam` optional charge penalty per residue (0.0 in the runs).

- Output `design.json`: ordering, per-step spacer counts, sequence, budget check.
- Reproducible structural counts across the three shells (all 132 ordered pairs connectable, path through 12/12 poses in each):

	shell 1	shell 2	shell 3
sequence	RGGDGGKGGGGLGGGKGIGEGWGGDGS GGEGS	RGEGGEGGKGGFGGDGGIGLGGSGWGGGGLGGS	YGGDGF GKGGGLGGGK
residues	33	33	34
spacers used / budget	21 / 16	21 / 16	22 / 16
worst closure	0.196 Å	0.120 Å	0.188 Å

8. Linker substitution

- `facebook/esm2_t12_35M_UR50D`; glycine spacers masked and refilled, designed positions untouched.
- **Sampled, not argmax**: candidates with probability $\geq$ `--prob-cutoff 0.05`, `--seed 1`, `--variants 2` per design → `esm1`, `esm2` per shell.
- Net charge is not controlled and moves a lot — record it per variant, since it drives the artifact in §13:
 - shell 1: design `-1`; shell 2 `esm1 0`, `esm2 +5`; shell 3 `esm1 0`, `esm2 +4`.

9. Null control

- `shuffle_control.py`: randomise which side chain occupies which slot, **keeping the glycine linker pattern**, `--seed 2`, plus one ESM2-filled variant (`shuffle_control_esm0`).
- Matched to the design on composition, length, 22 glycines, spacer pattern and net charge (0); **0 of 12 side chains left in their designed slot**.
- Rationale to state: shuffling the whole string instead produces 4–5 adjacent side-chain pairs with no glycine between them, randomising spacing as well as ordering — so the control tests **ordering only**.
- No fragment IE is fabricated for the control (it has no placed poses); `sequences.csv` untouched.

10. Co-folding

- **Boltz-2 2.2.1**, one peptide chain + ligand, `msa: empty`.
- Contact constraints written from the designed shell: ligand atom ↔ residue index, `max_distance` = contact cutoff `--cutoff 4.5 Å` + `--margin 1.0 Å`; the `--top N` tightest designed contacts.
- **Both flags are required and this must be in Methods**:
 - `force: false` constraints are **discarded by the featurizer**, not softened;
 - `force: true` alone is conditioning — the sampling-time guidance is gated behind `--use_potentials`, off by default. Runs used `--force and --boltz-args=--use_potentials`.
- Constraint ladder per sequence: **control (none), f4, f8, f12** → 12 structures per shell, 3 shells = **36 structures**.
- Affinity head omitted: Boltz 2.2.1 calls `AllChem.Descriptors.MolWt`, which RDKit 2026.03.6 no longer exposes. Independent reason to omit: it spans only 1.65 kcal/mol across 37 sequences and scored a complex with the ligand 6.6 Å away confidently.
- Exact command (`boltz_env.run_boltz`): `boltz predict <yaml> --num_workers 0 --out_dir <dir>`, plus `--use_potentials` on the forced runs. Nothing else is overridden, so the sampling parameters are the 2.2.1 defaults, enumerated from the pinned install (`boltz_local/.venv/.../boltz/main.py:1050-1056, 1231`):

parameter	value	note
model	<code>boltz2</code>	

recycling steps	3	
sampling steps	200	
diffusion samples	1	one structure per input, hence <code>model_0</code>
step scale	1.5	Boltz-2 default (1.638 is Boltz-1)
max parallel samples	None	
output format	mmCIF	
num workers	0	the one non-default, passed explicitly
seed	None — no seeding	
affinity head	not requested	see the RDKit incompatibility above

- Hardware and precision, from the fold logs: **MPS**, bfloat16 automatic mixed precision, on all **162** logged folds (`grep "GPU available" runs/octinoxate/boltz/*.log → 162 × (mps), used: True`). No fold in the project ran on CUDA or CPU.
- Structure taken: `*_model_0.cif` (the only sample, given `diffusion_samples 1`).
- **Consequence to state plainly:** with no seed and `diffusion_samples 1`, **the folds are not bit-reproducible** — re-running a sequence gives a different sample from the same distribution. Every structure in the paper is one draw. **[decide]** whether to say this in Methods (it is honest and cheap) or to re-fold a subset with `--seed` to quantify the spread. The latter is the referee-proof option and costs 2–3 min per fold.

11. Structural metrics (`check_fold.py`)

- `enclosed` — fraction of **200** sampled directions out of the ligand centroid that run into peptide; each ray given a 3 Å tolerance out to 12 Å. *Is the ligand in a shell or on a face.*
- `wrapped` — fraction of ligand heavy atoms with a peptide heavy atom within `--wrap 4.5 Å`. *Contact census.*
- `engaged` — count of the 20 ligand heavy atoms in contact (`--contact 4.0 Å`).
- `centroid_sep` — ligand-to-peptide centroid distance, to be read against `peptide_rg`.
- `peptide_rg` — radius of gyration. Empirically the cleanest discriminator of the design objective: encapsulating folds 9.3–9.8 Å vs groove binders 11.8–14.9 Å, no overlap, where `enclosed` separates the same two groups by 0.015 and `wrapped` inverts them.
- `helical_fraction`, `beta_fraction` — ϕ/ψ basin counts with deliberately generous windows (helix $\phi -63 \pm 35$, $\psi -43 \pm 35$) because Boltz geometry is unrefined; termini excluded.
- `helical_hbonds` / `nonlocal_hbonds` — backbone N...O < 3.3 Å split at sequence separation 3–5 (i,i+4) vs >5 (strand pairing). **The split is the method point:** 81% β with 6 non-local bonds is an extended unpaired chain, not a sheet; a combined count ranks a pure helix highest.
- `hints` — requested contacts satisfied / requested. Report it; do not treat it as success (uncorrelated with everything else measured).
- Overlay to the designed shell (`overlay.py`): superposition on the ligand, RMSD 0.06–0.85 Å, so the designed-position comparison is sound; designed position counted reproduced within 3 Å.

12. Static energetics

- Hydrogens added to Boltz's heavy-atom CIFs with **pdifix**, then **relaxed with heavy atoms fixed** (`relax_hydrogens, fmax 0.10 eV/Å, steps 75`).
 - Justification to state: `pdifix`'s placement is **not deterministic** — same structure twice gave –14.47 and –13.31 kcal/mol (1.2 kcal/mol from hydrogens alone); three protonations of one CIF differed by up to 2.3 Å, RMSD 0.78 Å, heavy atoms untouched. Relaxing hydrogens brings reproducibility to **0.08 kcal/mol**. Stable on a Linux container; unstable on macOS.
- `interaction = E(complex) - E(peptide) - E(ligand)`, all three **at the complex geometry** → rigid interaction energy, contains no strain.
- `ligand_strain = E(bound ligand, exactly as the complex relaxation left it) - E_ref`, **a single point**, no re-relaxation of any kind.
- `E_ref` = one shared reference per ligand (`ligand_reference.py`): 20 ETKDGV3 embeddings from the run's own

SMILES (`--seed 0xC0FFEE`), MMFF94 optimised (`maxIters 2000`) and ranked, the **5 lowest** relaxed with UMA at `fmax 0.01 eV/Å`; lowest kept. Value used throughout: **-557162.371 kcal/mol-equiv**. `strain_global.py` refuses to run if the stored reference's SMILES or model does not match the run.

- Why one shared reference and not a per-structure relaxation — **one sentence in Methods, numbers in SI**: relaxing each complex's own bound pose measures every structure against a different local minimum, and the term then moves with the relaxation settings (+1.5 to +7.7 kcal/mol between `fmax 0.1` and `0.01`) and with `pdffixer`'s non-deterministic hydrogens (3.6 kcal/mol on bit-identical heavy atoms) rather than with the bound geometry.
- **The bound state is not re-relaxed, in whole or in part** — including no re-placement of hydrogens on the isolated ligand, which would insert an intermediate state and report only the second leg.
- Negative strain = failed reference (a bound pose below the supposed global minimum), not a finding; re-run the conformer search with more embeddings.
- Terms deliberately **not** reported, each disabled by default in the code, with the reason:
 - peptide strain (`--terms strain_peptide`) — a free peptide relaxed in vacuum collapses into a hydrogen-bonded ball: 237.8 kcal/mol on a 33-mer, not converged. Needs implicit solvent, which puts it in MM/GBSA territory.
 - explicit-water cavity desolvation — ~420 atoms, still descending 1.6 kcal/mol per step at step 102 of 200, so the number would be set by the step budget rather than the physics.
- Known biases of the static energy, to state in Methods rather than defend in Results:
 - grows with system size (Spearman **-0.52** against peptide length) → compare matched lengths;
 - rewards net charge: across 36 folds, interaction energy correlates **-0.56** with charged- residue fraction;
 - a **positive** interaction energy means a broken structure — check heavy-atom contacts under 2.6 Å. Of three found here, one was an $n \rightarrow \pi^*$ approach 70° out of plane, one a CH...O bond 17° off the C-H vector, one genuinely unphysical (carbonyl O 2.55 Å from an ether O, no donor).

13. Molecular dynamics

- Structure split for MD (`cif_to_md.py`): Boltz writes peptide + ligand in one heavy-atom CIF; the ligand SDF is built by **assigning bond orders from the run's SMILES as a template** before adding hydrogens — without it the aromatic ring and the ester come out as single bonds.
- Force fields: **ff14SB** (`amber14/protein.ff14SB.xml`), **TIP3P** (`amber14/tip3p.xml`, ions bundled), ligand **GAFF-2.11** via `openmmforcefields` `GAFFTemplateGenerator`.
- Solvation: padding **1.5 nm, rhombic dodecahedron** box (`--box-shape dodecahedron`), 0.15 M ionic strength, neutralised, pH 7.0 protonation by `pdffixer` (`prep-protein`).
 - Box shape is a methods point worth one line: same minimum-image distance in ~71% of the volume — 33,045 → **23,121** particles for the rod, 30% saving, no change in physics.
- System: nonbonded cutoff **1.0 nm**, PME, **HBonds** constraints, rigid water.
- Protocol: L-BFGS minimisation to a **10 kJ/mol/nm** force tolerance (no iteration cap); **5,000 steps** restrained equilibration with **100 kJ/mol/nm²** on protein heavy atoms; production **10,000,000 steps × 2 fs = 20 ns**, Langevin middle integrator, **300 K**, friction **1 ps⁻¹**, MonteCarlo barostat **1 atm** every **25 steps**; frames every 500 steps.
- Platforms: local Apple Silicon **OpenCL** (single precision; the restrained equilibration phase has to run on CPU there) vs rented **A10G CUDA** via `modal_md.py`, which imports `openmm_md.dynamics.run` so remote and local trajectories come from identical code.
 - Rates: 1.72 ms/step at 8,647 particles (local) vs 0.376 ms/step at 23,121 (A10G); 20 ns of the 23k system ≈14 h local vs **63 min**.
 - Trap worth a footnote: the `auto` platform probe does not list CUDA and silently selects CPU on a GPU host; CUDA passed explicitly. Warm-up required before timing (lazy compile/autotune reads 2–5× too slow).
- Six trajectories, 20 ns each: `orig_f12`, `s3_orig_f12`, `s3_esm2_f4`, `s2_esm2_control` (designs) and `shuffle_control`, `shuffle_control_esm0` (nulls).
- **[decide]** whether to report the frame spacing difference explicitly: `orig_f12` was saved at 1 ps, the rest at 10 ps,

and release statistics are resampled to a common spacing (see §15).

14. MM/GBSA

- **MMPBSA.py** (AmberTools 24.8), **single-trajectory** binding mode with pre-stripped receptor/ligand prmtops (`cp/-rp/-lp`).
- **GBn2**, **igb = 8**, saltcon = 0.15 M, mbondi2 radii set explicitly (ParmEd leaves RADII at 0 when converting from OpenMM, which would give a NaN GB term).
- Every frame used: `startframe=1`, `endframe=N`, `interval=1` over the wrapped solute-only trajectory (`analyze` writes `traj_wrapped.xtc/.pdb`; it is a required step, not optional).
- Parameters reused from the production system, so MM/GBSA and MD are the same force field.
- **No entropy term**; single-trajectory means BOND/ANGLE/DIHED are exactly zero by construction — so ΔG contains **no ligand strain**, which is why interaction terms and not the static sum are the fair comparison against UMA.
- Convergence test to report as methods, on `orig_f12` (leading n ns of one 20 ns run): -25.37 ± 0.50 (40 ps pilot), -21.52 (5 ns), -16.86 (10 ns), -14.61 (15 ns), **-13.71 ± 0.04** (20 ns); successive changes 3.85, 4.66, 2.25, 0.90 — halving. Point to make: a short window gives a **confidently wrong** answer, not a noisy one (the pilot's ± 0.50 is sampling error over 40 correlated frames of the starting pose and overstates binding by 11.7 kcal/mol).

15. Trajectory analysis

- `md_contacts.py`: heavy-atom contacts at `--cutoff 4.0 Å` (matched to `fold_check.csv` so counts are comparable), centroid separation, peptide Rg, residence within 10 Å, release episodes, in `--deciles 10` windows.
 - **Release statistics resampled to a common** `--resample-ps 10.0` spacing — otherwise a finely saved run looks worse. Demonstration: `orig_f12` shows 8 released frames at 1 ps and **zero** at 10 ps, all eight being isolated single frames.
- `pair_contacts.py`: how many designed side-chain **pairs** contact the ligand simultaneously. Slots taken from the parent design's non-glycine positions, so shells, ESM variants and shuffles are handled identically. `--cutoff 4.0 Å`, `--persist 0.05` (pair held $\geq 5\%$ of the run).
 - Framing to state in Methods, since it defines what the construction delivers: n slots give $n(n-1)/2$ pairwise options, chain connectivity hands over the $n-1$ adjacent pairs free, so the informative band is **$[n-1, n(n-1)/2]$** = [11, 66] at twelve slots; the informative content is the non-adjacent pairs.
 - Derived measures: mean side chains engaged, % frames with ≥ 2 engaged, pairs realised, band position, pairs held $\geq 5\%$, long-range pairs (gap ≥ 7), dead slots.
- `md_frames.py`: representative frames for figures — ten-model ensemble superposed on the peptide and the **medoid** (frame nearest the window mean) of the last `--last-fraction 0.10` of the run, or an explicit `--window-ns` interval.
 - Disclosure that belongs in a caption, not hidden: `orig_f12` is shown at 11.0 ns rather than at the end, because its last 2 ns are the only stretch where the ligand is back in the cavity.

16. Statistics and reporting conventions

- Correlations by `correlate.py`, joining the geometry, energy and overlay CSVs on structure name; Pearson and Spearman reported with n beside each.
- **--group is mandatory practice, not a convenience**: these correlations invert between binding mechanisms. Documented instance — interaction energy vs centroid separation runs $+0.23 / +0.70 / +0.93$ at net charge $-1 / 0 / +5$, and vs enclosure $-0.56 / -0.76 / +0.18$; pooling all twelve reports -0.14 and $+0.42$, hiding both.
- n is small everywhere and should be said so once, in Methods: four structures per sequence, twelve per shell, 36 static structures, **six** trajectories.
- Spearman ρ at $n = 5-6$ quoted as indicative; the best predictor found (mean simultaneous engagement) fell **$+0.90 \rightarrow +0.83$** when the sixth structure was added and got that structure's pair backwards. **[decide]** whether to quote both columns as the README does.

17. Software and versions

component	version	role
Python	3.12	design pipeline
fairchem-core (UMA <code>uma-s-1p2p1</code>)	2.22.0	every static energy
ASE	3.29.0	BFGS, structure handling
RDKit	2026.03.6	SMILES, ETKDGV3, MMFF94, bond perception
PyTorch	2.13.0	UMA, ESM2
transformers (ESM2 <code>esm2_t12_35M_UR50D</code>)	5.17.0	linker substitution
pdbfixer / OpenMM (design env)	1.12.0 / 8.6.1	hydrogen addition
SciPy	1.18.1	L-BFGS-B closure search, Held–Karp support
Boltz	2.2.1	co-folding with contact constraints
OpenMM (MD env)	8.4.0	dynamics
openmmforcefields / openff-toolkit	0.15.1 / 0.18.0	GAFF2 ligand parameterisation
MDTraj / ParmEd	1.11.1 / 4.3.1	trajectory and topology handling
AmberTools (MMPBSA.py)	24.8	MM/GBSA

- Note for the record: **FAIRChem ≥ 2.22 refuses the UMA 1.0 checkpoint**, directing to `fairchem-core≤2.21.0`; `uma-s-1p2p1` is used throughout.
- Two environments, called at their boundary: the design pipeline is pip-installable; GAFF2/OpenFF parameterisation is not, so MD/MM-GBSA lives in a conda environment (`openmm-md`).

18. Reproducibility and data

- Seeds fixed and recorded: placement `--seed`, closure sweep 0, ETKDG `0xC0FFEE`, ESM2 fill 1, shuffle 2.
- Every relaxed geometry is written, not derived and discarded (`boltz/structures/`), because each costs 45–60 min; per-term results are checkpointed to `partial_*.json` as computed.
- UMA runs logged separately (`uma_logs/`) from non-UMA runs (`logs/`): step counts, timings and convergence are the only record of how an energy was arrived at.
- **[decide]** data availability statement: repository URL, and whether trajectories (`md/<structure>/prod*`) or only the stripped solute + `mmgbsa_summary.csv` are deposited.

Ordering suggestion for the section

1. Ligand and fragment library (§1–2) — short.
2. Placement and scoring (§3), shell selection (§4).
3. Reachability and the budget (§5–6), assignment (§7) — **this is the novel method; it deserves the most space and the geometry figure.**
4. Sequence construction: linkers (§8), controls (§9).
5. Co-folding and constraints (§10).
6. Metrics (§11) and static energetics (§12) — the strain-reference argument (§12) is long; it may belong in SI with a two-sentence pointer.
7. MD and MM/GBSA (§13–14), trajectory analysis (§15).
8. Statistics (§16), software (§17), availability (§18).

Candidates for SI rather than Methods

- The fmax sweep and the pdbfixer determinism measurements behind §12's two justification bullets.
- The MM/GBSA convergence table (§14) — arguably Results, since "run 20 ns, do not trust 5" is a finding about the protocol.
- Platform/timing table (§13) — SI, unless the paper makes a cost argument.

Scope note

This outline documents the pipeline as it currently stands. The superseded centroid-linker construction (`build_sequence's round(d / 3.8) - 1`), its twenty-odd designs, the `--linker-slack` sweep, the cyclic variants, the random-sequence control and the abandoned per-structure strain reference are deliberately absent: they belong in the Introduction as motivation or nowhere, not in Methods. The measured reachability statistics that *justify* current choices (§5's 0/810 at $k = 0$, §6's budget) are kept, since they are properties of the method being described.

Gaps to close before the section is complete

- **[gap]** Shell 3's pose overlap with shells 1 and 2 is described qualitatively but never counted; shell 2 vs shell 1 is 5/12.
- **[gap]** Shell 1's twelve folds were never scored for reproduced designed positions (`boltz/overlay.csv` holds sequence-level comparisons only), so that column is blank, not zero — say so or fill it.